Expression, Purification, and Functional Analysis of a Y255A Mutant of Catechol 2,3-Dioxygenase

Word count: 5557

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1. Abstract

Catechol 2,3-dioxygenase (EC 1.13.11.2) is a non-heme iron-dependent enzyme encoded by the *xylE* gene of the TOL plasmid pWW0 from *Pseudomonas putida* mt-2. This enzyme catalyses the extradiol ring cleavage of catechol to form the yellow product 2-hydroxymuconate semialdehyde and plays a key role in the microbial degradation of aromatic compounds. In this study, 6xHis-tagged catechol 2,3-dioxygenase was expressed from the plasmid pHISCAT in *Escherichia coli* under IPTG induction and purified using nickel-nitrilotriacetic acid (Ni-NTA) affinity chromatography. The identity of pHISCAT was confirmed by restriction enzyme digestion, and recombinant protein expression was demonstrated by SDS-PAGE, which revealed a band at the expected molecular weight of approximately 37 kDa. The purified enzyme displayed a specific activity of approximately 3.2 U/mg. Site-directed mutagenesis was performed using the QuikChange II method to introduce a Y255A (tyrosine to alanine) substitution at position 255, a conserved active site residue proposed to participate in substrate deprotonation. The mutation was confirmed by Sanger DNA sequencing. Qualitative assessment of enzyme activity by catechol spray assay showed that colonies containing the Y255A mutant displayed reduced yellow colouration compared to wild-type controls, indicating that the mutation reduced but did not completely abolish catechol 2,3-dioxygenase activity. These findings partially support the hypothesis that Tyr255 contributes to the catalytic function of the enzyme, consistent with its proposed role in substrate ligand deprotonation at the Fe(II) active site.

2. Introduction

Aromatic compounds are widespread environmental pollutants, and numerous microorganisms have evolved enzymatic pathways to utilise these substances as carbon and energy sources. The TOL plasmid pWW0 of *Pseudomonas putida* mt-2 encodes the *meta*-cleavage pathway for the catabolism of monocyclic aromatic compounds, including toluene and xylenes, into intermediates that can enter the tricarboxylic acid cycle (Cerdan *et al.*, 1994). A key enzyme in this degradative pathway is catechol 2,3-dioxygenase (EC 1.13.11.2), also known as metapyrocatechase (MPC), which is encoded by the *xylE* gene. This enzyme catalyses the extradiol ring cleavage of catechol to produce 2-hydroxymuconate

semialdehyde, an intensely yellow-coloured product, with the incorporation of both atoms of molecular oxygen (Kita *et al.*, 1999). As such, the conversion of a colourless substrate to a brightly coloured product has made this enzyme a valuable tool in a range of biotechnological reporter assays (Kita *et al.*, 1999).

Catechol 2,3-dioxygenase from *P. putida* mt-2 is a homotetramer comprising four identical subunits, each consisting of 307 amino acid residues with a molecular weight of approximately 35 kDa per subunit (Kobayashi *et al.*, 1995). Each subunit contains a single non-heme ferrous iron (Fe^{2+}) cofactor that is essential for catalytic activity. The crystal structure of the enzyme, resolved at 2.8 Å resolution, revealed that each subunit is organised into N-terminal and C-terminal domains of similar structure, with the active site located in a funnel-shaped cavity within the C-terminal domain (Kita *et al.*, 1999; Ishida *et al.*, 2002). The ferrous iron at the active site is directly coordinated by three conserved residues (His153, His214, and Glu265), forming a triad geometry characteristic of non-heme iron oxygenases (Kita *et al.*, 1999). Notably, the active site iron is susceptible to oxidation to the ferric state, leading to enzyme inactivation, and acetone has been shown to act as both a protective agent and a competitive inhibitor against catechol (Kobayashi *et al.*, 1995). The enzyme displays broad substrate specificity, catalysing the extradiol cleavage of catechol and several substituted catechols, including 3-methylcatechol and 4-methylcatechol, though the kinetic parameters vary depending on the substrate (Cerdan *et al.*, 1994).

Among the conserved active site residues, Tyr255 has been identified as potentially important for the catalytic mechanism. Ishida *et al.* (2002) proposed that Tyr255, together with His199, may play important roles in the deprotonation of substrate ligands and the formation of charge-transfer complexes at the Fe(II) centre. Understanding the contribution of individual residues to enzyme function can be achieved through site-directed mutagenesis, where a targeted amino acid is typically replaced with a small, uncharged residue like alanine in an approach known as an alanine scan. This strategy effectively removes the side chain functional group while minimally disrupting protein backbone conformation.

The *xylE* gene has been successfully expressed at high levels in *Escherichia coli*, producing recombinant enzyme with physical, chemical, and kinetic properties

identical to the native *P. putida* enzyme (Kobayashi *et al.*, 1995). *E. coli* is an effective host for recombinant protein production owing to its well-characterised genetics, rapid growth rate, and the availability of established expression systems (Rosano and Ceccarelli, 2014). The use of affinity tags, such as the hexahistidine (6xHis) tag, enables efficient purification of recombinant proteins via immobilised metal affinity chromatography (IMAC), where the polyhistidine tag binds selectively to nickel-nitrilotriacetic acid (Ni-NTA) agarose under native conditions (Wei *et al.*, 2010).

In the present study, the *xylE* gene was expressed as a 6xHis-tagged fusion protein from the plasmid pHISCAT in *E. coli*, and the recombinant catechol 2,3-dioxygenase was purified using Ni-NTA affinity chromatography. The identity of the plasmid was confirmed by restriction enzyme digestion, and the purified protein was characterised by SDS-PAGE, Western blotting, Bradford protein assay, and spectrophotometric enzyme activity assays measuring the formation of 2-hydroxymuconate semialdehyde at 375 nm. Additionally, site-directed mutagenesis was employed using the Agilent QuikChange II method to introduce a Y255A substitution (tyrosine to alanine at position 255) in catechol 2,3-dioxygenase, and the effect of this mutation on enzyme activity was assessed by catechol spray assay of induced *E. coli* colonies. It was hypothesised that the Y255A mutation would reduce or abolish catechol 2,3-dioxygenase activity, given the proposed involvement of Tyr255 in substrate deprotonation and charge-transfer complex formation at the active site (Ishida *et al.*, 2002).

3. Materials and Methods

3.1 Bacterial strains, plasmids, and growth conditions

Escherichia coli BL21 (DE3) cells containing the plasmid pHISCAT expressed 6xHis-tagged catechol 2,3-dioxygenase. pHISCAT (3617 bp) was constructed by cloning the *xylE* open reading frame from the TOL plasmid pWW0 of *P. putida* into the pHAT10 vector (Clontech Laboratories, USA) between the *Hind*III and *Bam*HI sites, with an in-frame N-terminal 6xHis tag under *lac* promoter control. XL1-Blue supercompetent cells (Agilent Technologies) were used for transformation of mutated plasmid DNA following site-directed mutagenesis. Cultures were routinely

grown in Luria-Bertani (LB) broth or on LB agar plates supplemented with 100 µg/ml ampicillin (LB-amp) at 37°C.

3.2 IPTG induction of recombinant protein expression

An overnight culture of *E. coli* containing pHISCAT was diluted and grown in 500 ml LB-amp at 37°C with shaking (200 rpm). The optical density at 600 nm (OD_{600}) was monitored until it reached approximately 0.5, at which point isopropyl-β-D-thiogalactopyranoside (IPTG) was added to a final concentration of 1 mM to induce expression of 6xHis-tagged catechol 2,3-dioxygenase. Aliquots of 1 ml culture were removed at 0, 1, 2, and 3 hours post-induction, and cell pellets were collected by centrifugation and stored at −20°C.

3.3 Restriction enzyme digestion and agarose gel electrophoresis

Restriction enzymes (Promega) were used per manufacturer's instructions (Promega Restriction Enzyme Resource). A typical reaction contained 250 ng plasmid DNA, 10 units of each restriction enzyme, and 2 µl of the appropriate 10× Promega restriction enzyme buffer in a total volume of 20 µl, incubated for 1 hour at 37°C. Double digests of pHISCAT were performed with *Bam*HI/*Hind*III, *Eco*RV/*Sac*I, and *Bam*HI/*Eco*RV to confirm plasmid identity. Agarose gel electrophoresis was performed according to Sambrook and Russell (2001). A typical gel contained 0.8% (w/v) agarose in 1×TAE buffer with 4 µl of 10,000× Biotium GelRed. Samples were loaded in 1× Promega Blue/Orange loading dye alongside an Invitrogen 1 kb Plus DNA ladder (500 ng), and gels were run at 100 V. Gel images were recorded using a gel imager.

3.4 SDS-PAGE and Western blotting

Cell pellets were lysed in 100 µl BugBuster Master Mix (Merck Millipore). Lysates (20 µl) were mixed 1:1 with 2× Laemmli sample buffer (Laemmli, 1970), heated at 95°C for 5 minutes, and 10 µl resolved on Mini-PROTEAN TGX precast gels (Bio-Rad) at 200 V in 1× Tris/glycine/SDS buffer. Bio-Rad Precision Plus Protein Kaleidoscope Standards and Invitrogen Benchmark His-tagged Protein Standards were included as molecular weight markers. Twin gels were run in parallel; one was stained with Bio-Safe Coomassie stain (Bio-Rad) for 1 hour, and the other was used for Western blotting. For the Western blot, proteins were transferred to a nitrocellulose

membrane in transfer buffer for 1 hour at 100 V using a Mini Trans-Blot system (Bio-Rad). The membrane was blocked with 3% (w/v) bovine serum albumin (BSA) in Tris-buffered saline (TBS), then incubated with Ni-NTA horseradish peroxidase (HRP) conjugate (Qiagen) to detect 6xHis-tagged proteins. Bands were visualised using HRP chromogenic substrate.

3.5 Purification of 6xHis-tagged catechol 2,3-dioxygenase

6xHis-tagged catechol 2,3-dioxygenase was purified under native conditions using Ni-NTA agarose affinity chromatography per the QIAexpressionist handbook (Qiagen). Cells from 50 ml induced culture were harvested, lysed in lysis buffer (50 mM NaH_2PO_4, 300 mM NaCl, 10 mM imidazole, pH 8.0), and the soluble fraction was applied to a gravity-flow Ni-NTA column. After three washes with wash buffer (20 mM imidazole), the bound protein was eluted in four sequential 1 ml fractions using elution buffer (250 mM imidazole). The four eluate fractions were pooled and concentrated using a Millipore Amicon Ultra-4 centrifugal filter unit (10 kDa molecular weight cut-off) at 4,000 × *g* and 4°C. Imidazole was removed by three rounds of buffer exchange with Buffer A (50 mM sodium phosphate, pH 7.5). Fractions from each purification step were analysed by SDS-PAGE as described above.

3.6 Protein concentration determination and enzyme activity assay

Protein concentration was determined by Bradford assay (Bradford, 1976) using the Bio-Rad Protein Assay reagent with BSA standards (Bio-Rad Protein Assay Instruction Manual). Serial dilutions of purified enzyme (1 in 5 to 1 in 3125, 10 µl each) were assayed in duplicate alongside BSA standards in 1 ml total volume containing 200 µl dye reagent. Absorbance was measured at 595 nm using a spectrophotometer. Catechol 2,3-dioxygenase activity was assayed spectrophotometrically at 375 nm by monitoring the formation of 2-hydroxymuconate semialdehyde (ε = 36,000 M^{-1} cm^{-1}) at 25°C (Kobayashi *et al.*, 1995). A typical 1 ml reaction contained 100 µl of 3 mM catechol, 850 µl Buffer A, and 50 µl of diluted enzyme (1 in 625). Absorbance readings were taken every minute for 10 minutes, and three replicate assays were performed. Specific activity was expressed as µmol/min/mg protein.

3.7 Site-directed mutagenesis, transformation, and screening

A Y255A (tyrosine to alanine) substitution was introduced into the *xylE* gene using the QuikChange II Site-Directed Mutagenesis Kit (Agilent Technologies) according to the manufacturer's instructions. Complementary mutagenic primers (Y255Afor: 5′-CTCACGGCAAGACCATC**GCC**TTCTTCGACCCGTCCGG-3′ and Y255Arev: 5′-CCGGACGGGTCGAAGAA**GGC**GATGGTCTTGCCGTGAG-3′; mutated codons in bold) were designed to change the codon at position 255 from TAC (Tyr) to GCC (Ala). After thermal cycling and DpnI digestion, mutated plasmid was transformed into XL1-Blue cells and plated on LB-amp agar. Eight colonies were sector-streaked onto LB-amp plates with nitrocellulose filters, induced on IPTG-containing plates, and screened by spraying with 100 mM catechol in a fume hood; yellow colour intensity was recorded at 0, 5, and 10 minutes. *E. coli* containing pHISCAT (positive control) and *E. coli* containing pHAT10 (negative control) were included. Plasmid DNA was extracted from six selected colonies using a commercial miniprep kit and digested with *Bam*HI and *Hind*III to confirm the presence of pHISCAT. One miniprep sample (92 ng/µl; $A_{260/280}$ = 2.08) was submitted for Sanger DNA sequencing to confirm the Y255A mutation.

3.8 Safety and ethical considerations

All experimental work was carried out in accordance with local safety regulations. A COSHH risk assessment was completed for catechol, which is classified as harmful by ingestion and skin contact. Catechol solutions were handled wearing nitrile gloves and safety glasses, and spraying of catechol was performed in a fume hood. *E. coli* strains used in this study are classified as Risk Group 1 organisms and were handled using standard microbiological aseptic technique. All biological waste was disposed of via autoclaving or into clinical waste containers as appropriate.

4. Results

4.1 Confirmation of pHISCAT plasmid identity

To confirm that the plasmid provided was pHISCAT, a plasmid map was generated from the DNA sequence data (Figure 1) and the plasmid was subjected to restriction enzyme digestion. The predicted total size of pHISCAT is 3617 bp, with key restriction sites at positions *Hind*III (140), *Cla*I (880), *Eco*RV (884), *Bam*HI (1089),

and *Sac*I (1108). Single digests with *Bam*HI, *Hind*III, and *Sac*I each produced a single linearised band migrating at approximately 3600 bp, consistent with the predicted plasmid size (Figure 2). The undigested control migrated faster than the linearised plasmid due to its supercoiled conformation. Double digests with *Bam*HI/*Hind*III produced two fragments of approximately 2700 bp and 950 bp, consistent with the predicted sizes of 2669 bp and 949 bp. Double digests with *Eco*RV/*Sac*I and *Bam*HI/*Eco*RV each produced a large fragment of approximately 3400 bp, though the corresponding small fragments (~224 bp and ~205 bp respectively) were faint or not clearly visible on the gel, as expected for very small DNA fragments at this agarose concentration. These restriction patterns confirmed that the plasmid was pHISCAT.

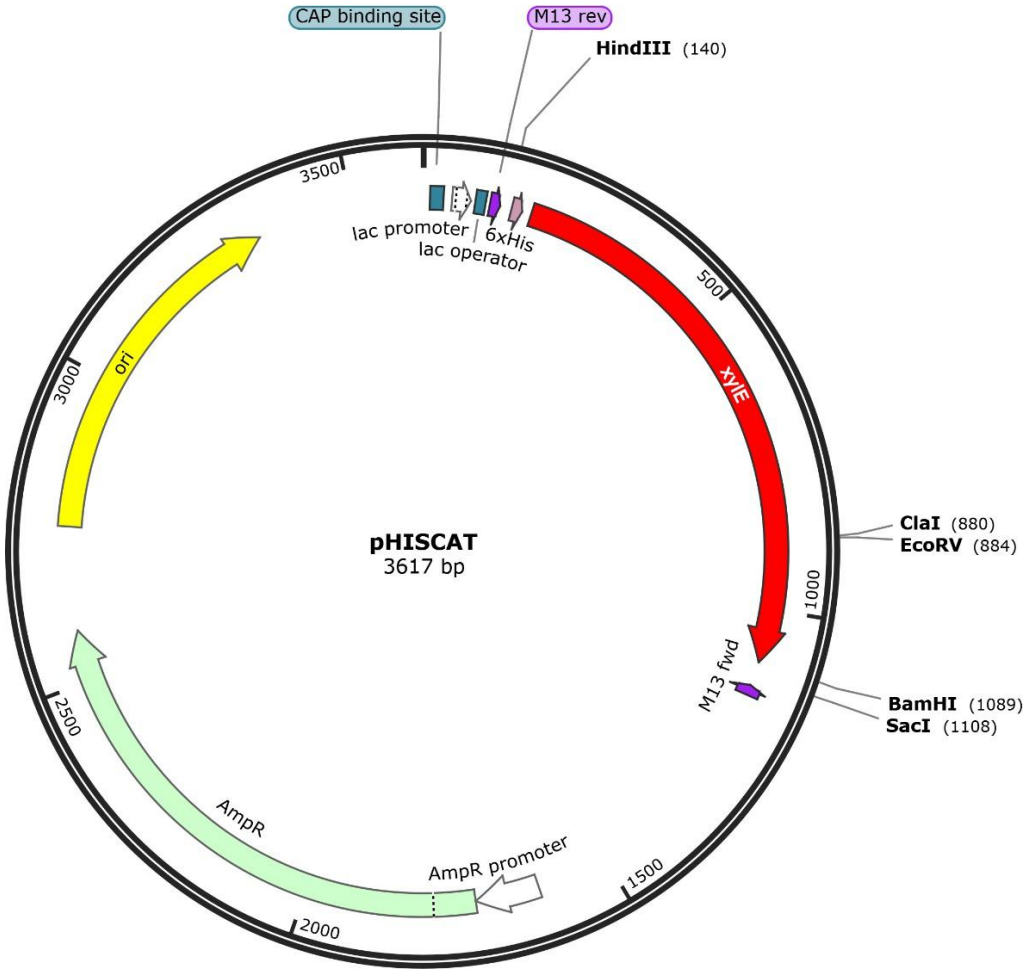


Figure 1. Plasmid map of pHISCAT (3617 bp) showing positions of key components including the *xylE* gene encoding catechol 2,3-dioxygenase, 6xHis tag, *lac* promoter/operator, CAP binding site, ampicillin resistance gene (AmpR), origin of replication (ori), and restriction enzyme sites for *Hind*III, *Cla*I, *Eco*RV, *Bam*HI, and *Sac*I. The plasmid was constructed by cloning the *xylE* open reading frame from the TOL plasmid pWW0 of *P. putida* into the pHAT10 vector (Clontech Laboratories).

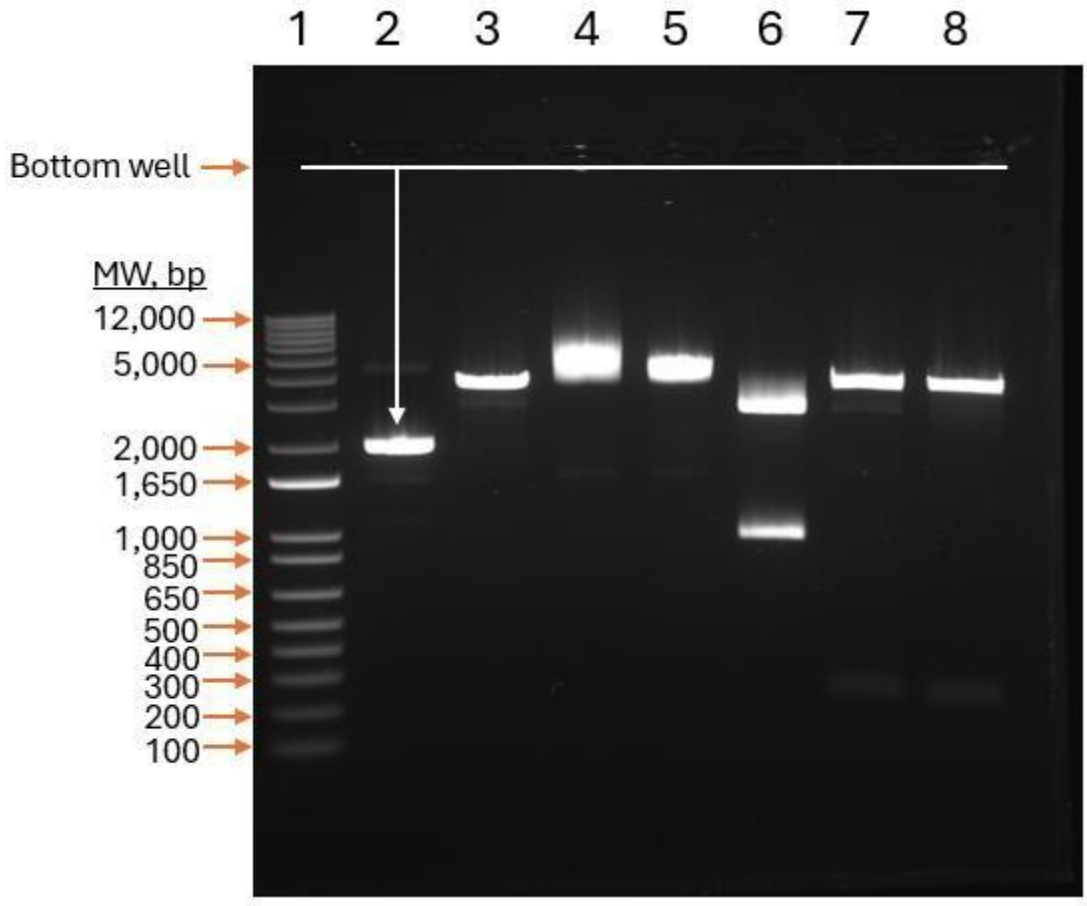


Figure 2. Agarose gel electrophoresis of restriction enzyme digests of pHISCAT plasmid. DNA fragments were separated on a 1×TAE, 0.8% (w/v) agarose gel containing GelRed. Lane 1 – Invitrogen 1 kb Plus DNA ladder (500 ng); Lane 2 – undigested pHISCAT; Lane 3 – pHISCAT digested with *Bam*HI; Lane 4 – pHISCAT digested with *Hind*III; Lane 5 – pHISCAT digested with *Sac*I; Lane 6 – pHISCAT digested with *Bam*HI and *Hind*III; Lane 7 – pHISCAT digested with *Eco*RV and *Sac*I; Lane 8 – pHISCAT digested with *Bam*HI and *Eco*RV.

4.2 IPTG-induced expression of 6xHis-tagged catechol 2,3-dioxygenase

To confirm expression of the recombinant 6xHis-tagged catechol 2,3-dioxygenase, *E. coli* cell lysates from the IPTG induction time course were analysed by SDS-PAGE (Figure 3). The predicted molecular weight of the 6xHis-tagged catechol 2,3-dioxygenase is approximately 37 kDa. A protein band at this position was visible in the 1-hour and 2-hour post-induction samples and appeared to increase in intensity relative to the background *E. coli* proteins, consistent with IPTG-induced expression. The 0-hour (pre-induction) sample showed the typical pattern of *E. coli* soluble proteins with no prominent band at 37 kDa. The 3-hour sample appeared anomalously faint across all protein bands, which was most likely due to a loading or cell lysis error rather than a genuine reduction in expression, since IPTG induction typically increases over the 3-hour period.

Western blot analysis of the same cell lysates using a Ni-NTA HRP conjugate to detect 6xHis-tagged proteins was unsuccessful, as no bands were detected on the membrane (Figure 4). This result may have been due to insufficient protein transfer from the gel to the nitrocellulose membrane, issues with the concentration or incubation time of the Ni-NTA HRP conjugate, or problems during the chromogenic detection step. Despite the unsuccessful Western blot, the SDS-PAGE results provided evidence of recombinant protein expression.

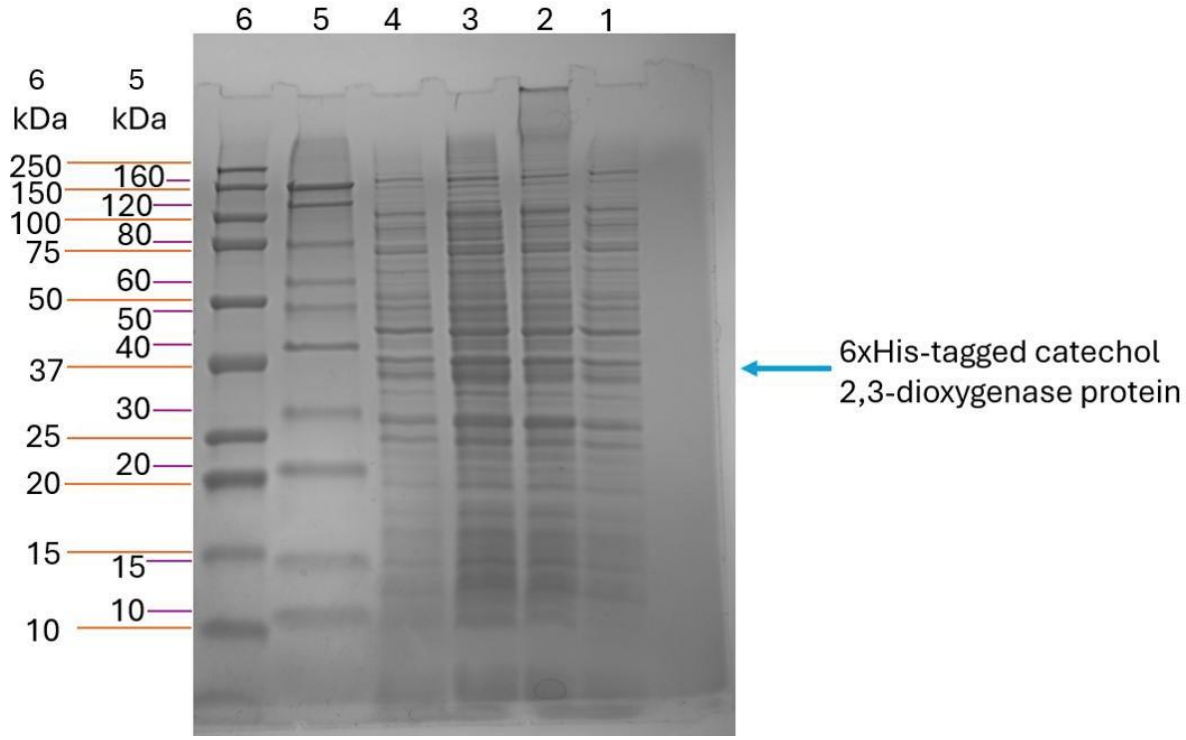


Figure 3. SDS-PAGE analysis of IPTG-induced expression of 6xHis-tagged catechol 2,3-dioxygenase in *E. coli*. Cell pellets from 1 ml culture at each time point were lysed in 100 µl BugBuster Master Mix. 10 µl of cell lysate was mixed with an equal volume of 2× Laemmli sample buffer and resolved on a Mini-PROTEAN TGX precast gel (Bio-Rad). Lane 1 – 0 h IPTG induction; Lane 2 – 1 h IPTG induction; Lane 3 – 2 h IPTG induction; Lane 4 – 3 h IPTG induction; Lane 5 – Invitrogen Benchmark His-tagged Protein Standard; Lane 6 – Bio-Rad Precision Plus Protein Kaleidoscope marker. The arrow indicates the expected position of the 6xHis-tagged catechol 2,3-dioxygenase at approximately 37 kDa.

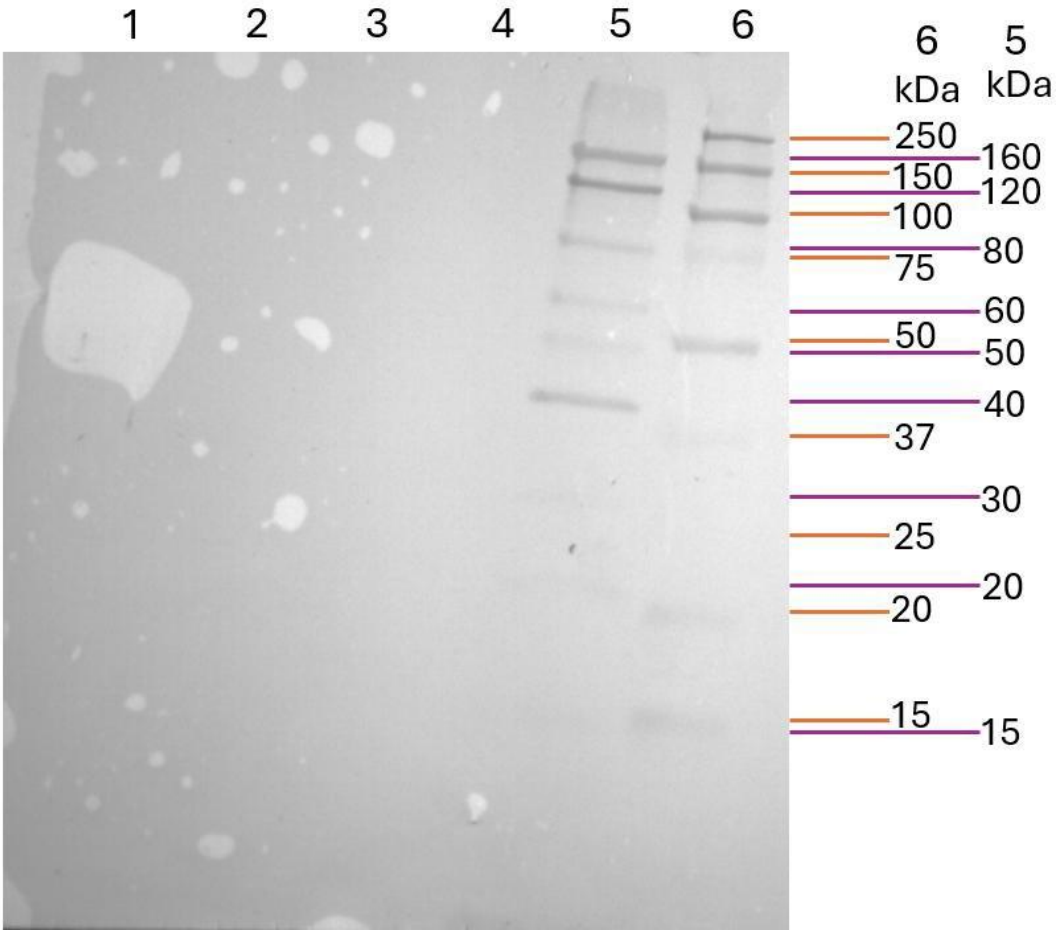


Figure 4. Western blot of *E. coli* cell lysates from the IPTG induction time course. Proteins were transferred to a nitrocellulose membrane and probed with Ni-NTA HRP conjugate to detect 6xHis-tagged proteins. Lane 5 – Invitrogen Benchmark His-tagged Protein Standard; Lane 6 – Bio-Rad Precision Plus Protein Kaleidoscope marker. No 6xHis-tagged catechol 2,3-dioxygenase bands were detected in the cell lysate lanes.

4.3 Purification of 6xHis-tagged catechol 2,3-dioxygenase

SDS-PAGE analysis of the Ni-NTA affinity purification fractions demonstrated successful isolation of the 6xHis-tagged catechol 2,3-dioxygenase (Figure 5). The cell debris fraction (Lane 1) contained a complex mixture of *E. coli* soluble proteins. The wash fractions (Lanes 2–4) showed progressively decreasing amounts of non-specifically bound proteins, with the third wash fraction being essentially clear, indicating that unbound proteins had been effectively removed. The four elution fractions (Lanes 5–8) each contained a prominent band at approximately 37 kDa,

corresponding to the expected size of the 6xHis-tagged catechol 2,3-dioxygenase. The first and second elution fractions also contained minor contaminating bands, likely representing *E. coli* metal-binding proteins that co-purify on Ni-NTA columns. The concentrated sample (Lane 9, 1 in 25 dilution) showed a predominantly single band at 37 kDa, demonstrating substantial enrichment of the target protein. The Bio-Rad Precision Plus Protein marker was loaded in Lane 10.

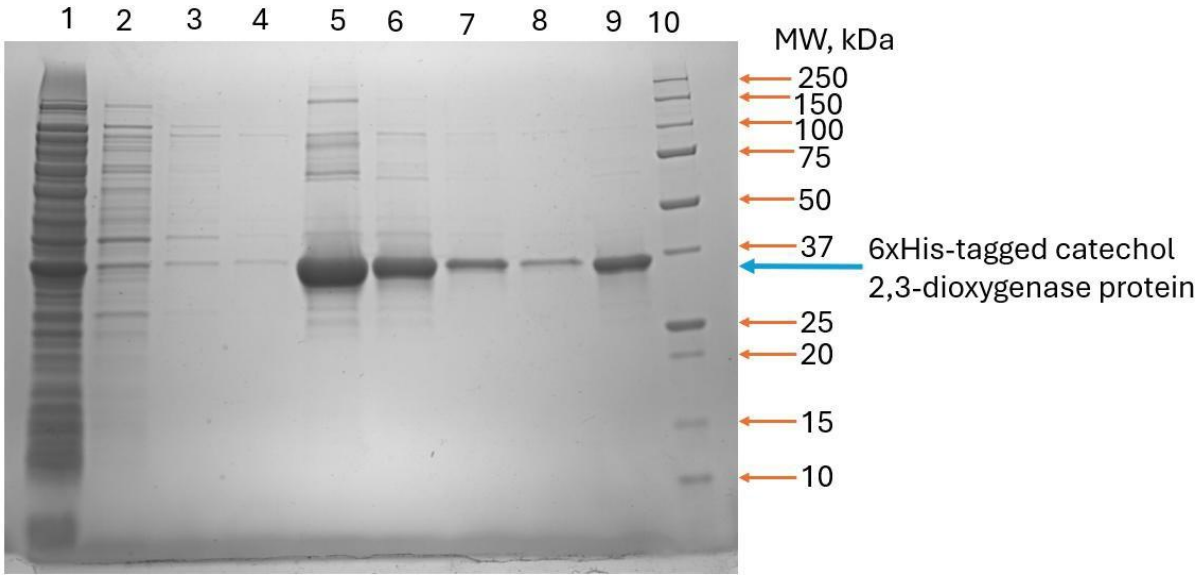


Figure 5. SDS-PAGE analysis of fractions from Ni-NTA affinity purification of 6xHis-tagged catechol 2,3-dioxygenase. Proteins were stained with Bio-Safe Coomassie stain (Bio-Rad). Lane 1 – cell debris; Lane 2 – 1st wash; Lane 3 – 2nd wash; Lane 4 – 3rd wash; Lane 5 – 1st elution; Lane 6 – 2nd elution; Lane 7 – 3rd elution; Lane 8 – 4th elution; Lane 9 – 1 in 25 dilution of concentrated purified 6xHis-tagged catechol 2,3-dioxygenase; Lane 10 – Bio-Rad Precision Plus Protein marker. The arrow indicates the 6xHis-tagged catechol 2,3-dioxygenase at approximately 37 kDa.

4.4 Protein concentration and enzyme activity

A BSA standard curve was generated using the Bradford assay (Figure 6, Appendix Table 4). The 1 in 25 dilution of purified 6xHis-tagged catechol 2,3-dioxygenase gave a mean absorbance at 595 nm of 0.212, which was the only dilution that fell within the measurable range of the standard curve (Appendix Table 5). The concentrated and 1 in 5 dilutions gave absorbance values above the standard curve range, while the 1 in 125, 1 in 625, and 1 in 3125 dilutions were below the detection limit. Only

160 µl of concentrated protein was recovered following the Amicon Ultra-4 concentration step, rather than the expected 400 µl, likely due to an additional centrifugation cycle that was required because the initial pooled eluate volume exceeded 4 ml.

Due to time constraints arising from the additional centrifugation step during protein concentration and the time required to set up the spectrophotometer, it was not possible to complete triplicate enzyme activity assays. The enzyme activity data presented (Figure 7, Appendix Table 6) are therefore representative values provided by the module academic with permission and are not the author's own experimental results. Three replicate assays of the 1 in 625 dilution of purified 6xHis-tagged catechol 2,3-dioxygenase were performed, measuring the formation of 2-hydroxymuconate semialdehyde at 375 nm over 10 minutes. All three assays displayed a characteristic enzyme kinetics profile, with a rapid initial rate of reaction that plateaued as the substrate became limiting (Figure 7). The initial rates were 0.176, 0.166, and 0.149 A_{375}/min for assays 1, 2, and 3 respectively. The decrease in initial rate from assay 1 to assay 3 was attributed to progressive inactivation of the enzyme upon dilution, a known property of catechol 2,3-dioxygenase (Kobayashi *et al.*, 1995). Using the Beer-Lambert law (ε = 36,000 M^{-1} cm^{-1}), the rates of 2-hydroxymuconate semialdehyde formation were calculated as 0.0048, 0.0046, and 0.0041 µmol/min respectively. The mean specific activity of the purified enzyme was calculated to be 3.2 µmol/min/mg protein (see Appendix Table 7 for the full calculation summary).

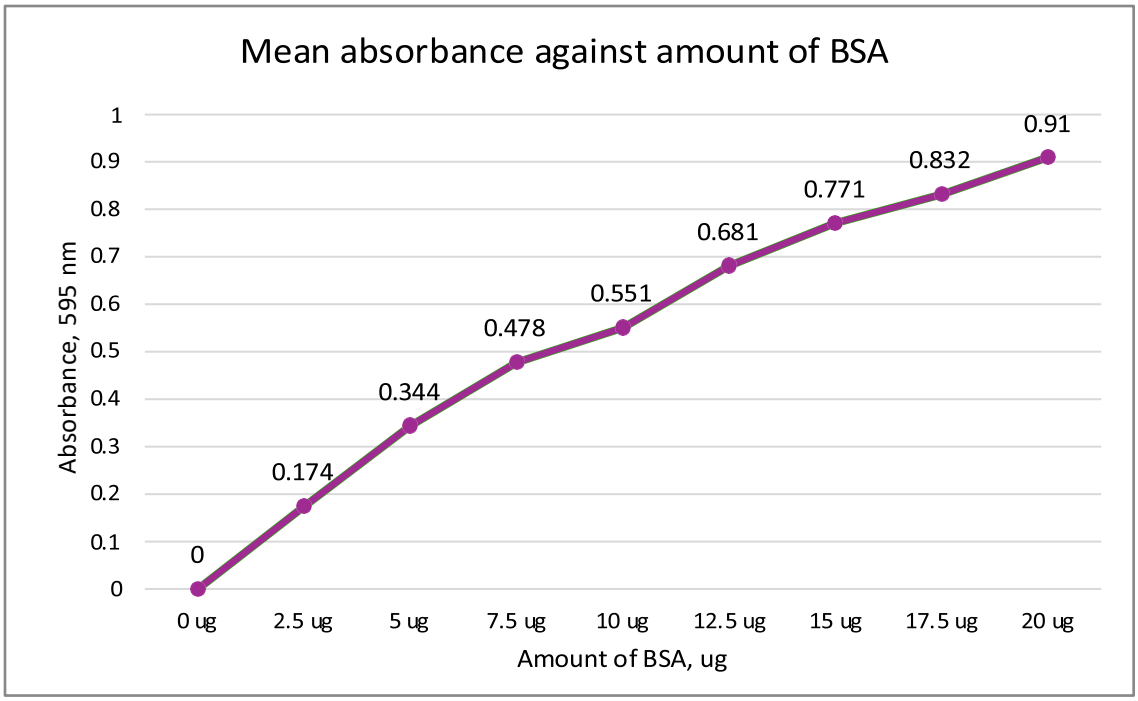


Figure 6. Standard curve of mean absorbance at 595 nm against BSA concentration (µg) for the Bradford protein assay. BSA standards ranging from 0 to 20 µg were assayed in duplicate using the Bio-Rad Protein Assay dye reagent.

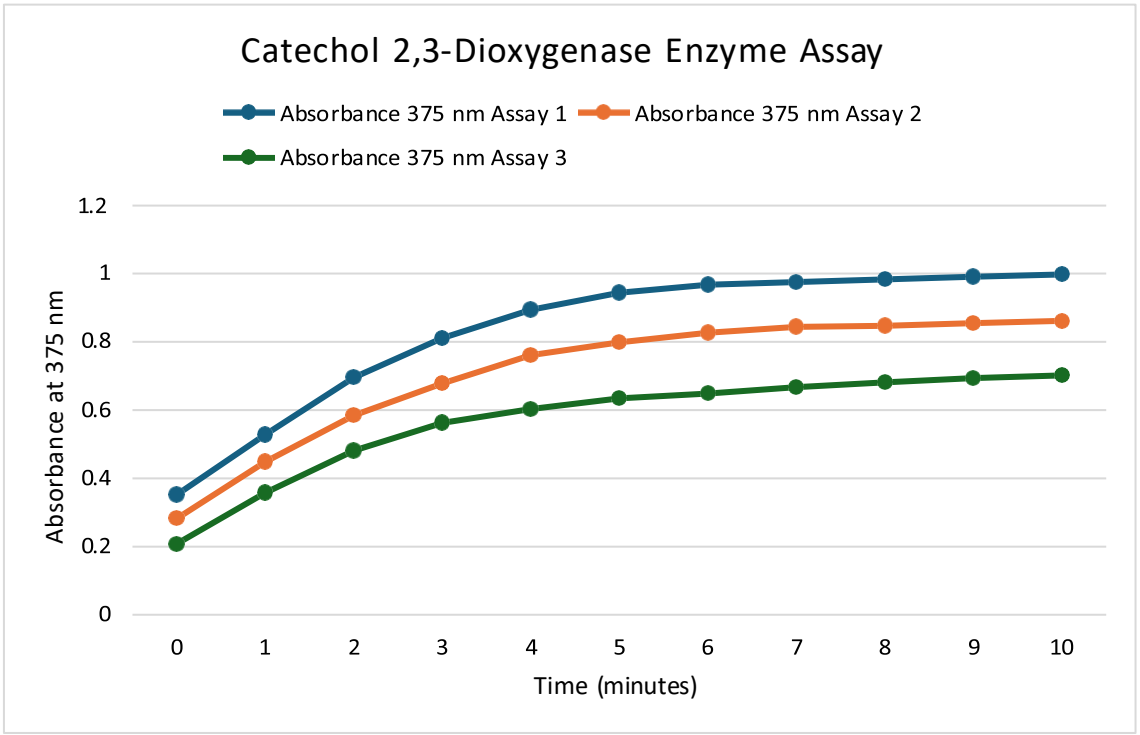


Figure 7. Catechol 2,3-dioxygenase enzyme activity assay. Absorbance at 375 nm was measured over 10 minutes for three replicate assays of purified 6xHis-tagged catechol 2,3-dioxygenase (1 in 625 dilution). Each 1 ml reaction contained 100 µl of 3 mM catechol, 850 µl Buffer A, and 50 µl of enzyme. Note: these data are representative results provided by the module academic with permission, not the author's own experimental data (see text for explanation).

4.5 Site-directed mutagenesis of catechol 2,3-dioxygenase (Y255A)

Following site-directed mutagenesis and transformation, colonies were obtained on all six LB-amp plates, with counts of 70, 79, 81, and 56 on the 100 µl plates and 4 and 21 on the 20 µl plates (Table 1). The class control experiment demonstrated that DpnI digestion effectively removed parental methylated DNA: the pWhitescript control without DpnI produced too many colonies to count on the 100 µl plates, while pWhitescript with DpnI treatment produced only 8 and 12 blue colonies on the 100 µl plates and 0 white colonies, indicating efficient digestion of the parental template (Table 2).

Eight colonies were selected for screening by catechol spray assay (Table 3, Figure 8). The *E. coli* pHISCAT positive control showed strong yellow colouration (+++) within 5 minutes of spraying, confirming functional wild-type catechol 2,3-dioxygenase activity. The *E. coli* pHAT10 negative control showed no yellow colouration at any time point, as expected for cells lacking the *xylE* gene. Sectors 5 and 6 showed no growth on the LB-amp agar plates and were therefore not assessable. Of the six remaining sectors, sectors 1 to 4 displayed reduced yellow colouration (+ to ++) compared to the wild-type pHISCAT control at 10 minutes, suggesting that the Y255A mutation reduced but did not completely abolish catechol 2,3-dioxygenase activity. Sectors 7 and 8 showed stronger yellow colouration (++ to +++) approaching wild-type levels, which may represent colonies in which the mutagenesis was unsuccessful and the plasmid retained the wild-type sequence.

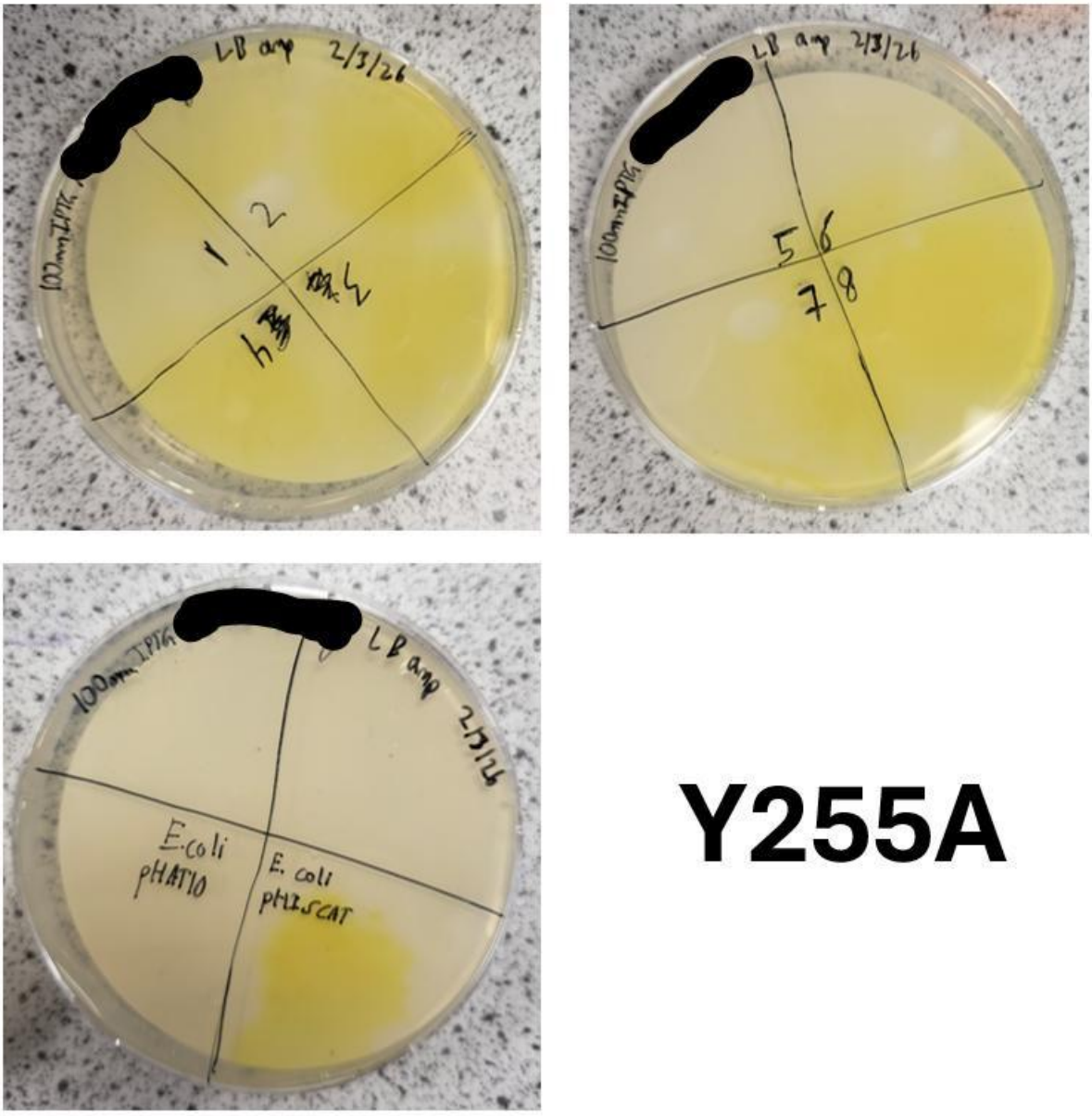


Figure 8. Catechol spray assay of IPTG-induced *E. coli* sector streaks for qualitative assessment of catechol 2,3-dioxygenase activity. Sectors were sprayed with 100 mM catechol and photographed after 10 minutes. Sectors 1–4 and 7–8 contain *E. coli* colonies with potentially mutated (Y255A) pHISCAT. Sectors 5 and 6 showed no growth. Controls: *E. coli* pHAT10 (negative control, no *xylE* gene); *E. coli* pHISCAT (positive control, wild-type catechol 2,3-dioxygenase). Yellow colouration indicates conversion of catechol to 2-hydroxymuconate semialdehyde.

4.6 Confirmation of mutation by restriction digestion and DNA sequencing

Plasmid DNA was extracted from six of the selected sectors by miniprep and digested with *Bam*HI and *Hind*III. All six miniprep samples produced two fragments at approximately 2669 bp and 949 bp (Figure 9), consistent with the expected restriction pattern of pHISCAT and confirming that all colonies contained the plasmid. One miniprep sample was selected for Sanger DNA sequencing based on restriction digest quality and DNA concentration (92 ng/µl; $A_{260/280}$ = 2.08). The sequencing data confirmed the successful introduction of the Y255A mutation, showing a TAC to GCC codon change at position 255 (Figure 10). No other mutations were detected in the sequenced region of the *xylE* gene, confirming that a clean, single-site Y255A substitution had been introduced.

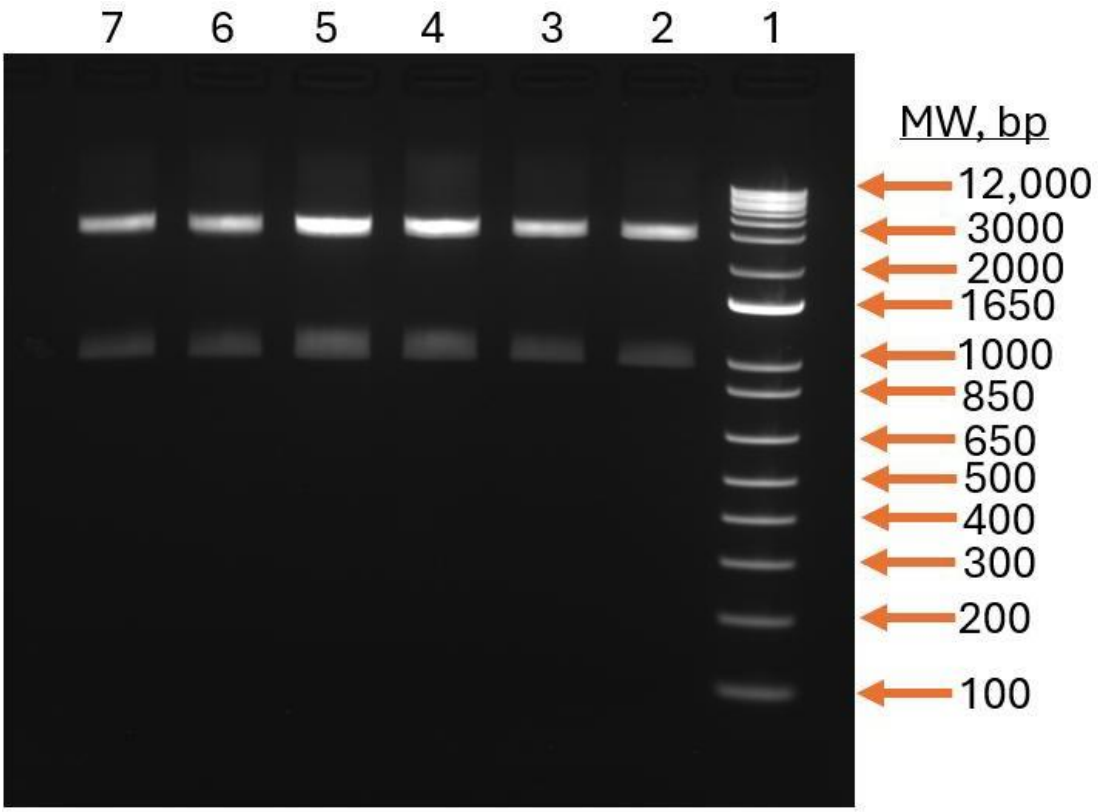


Figure 9. Agarose gel electrophoresis of miniprep plasmid DNA digested with *Bam*HI and *Hind*III. DNA fragments were separated on a 1×TAE, 0.8% (w/v) agarose gel containing GelRed. Lane 1 – Invitrogen 1 kb Plus DNA ladder (500 ng); Lanes 2–7 – miniprep DNA from six selected mutant sectors digested with *Bam*HI and *Hind*III. Two fragments of approximately 2669 bp and 949 bp are visible in each lane, consistent with the expected restriction pattern of pHISCAT.

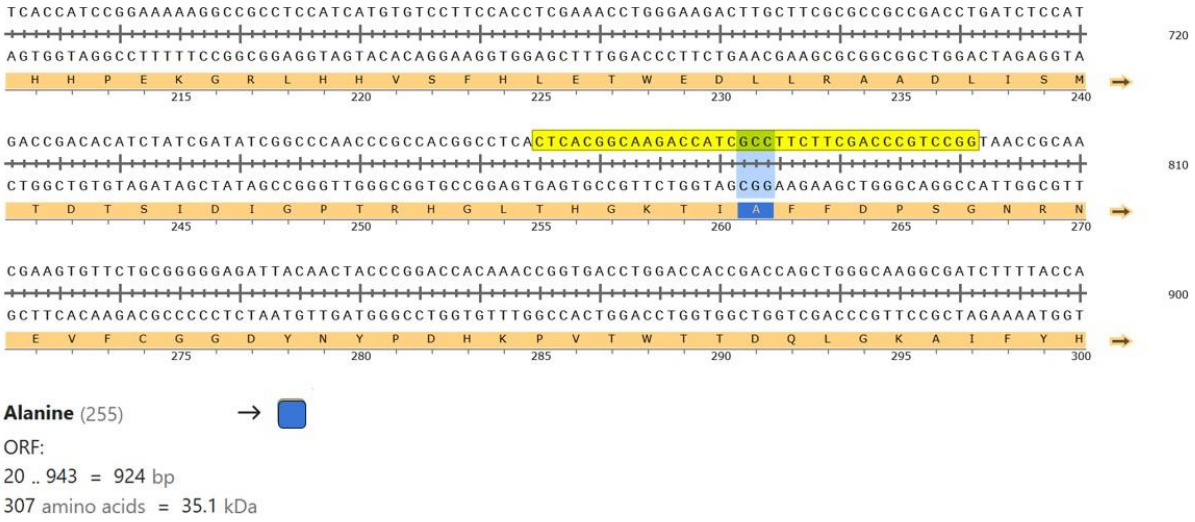


Figure 10. Sanger DNA sequencing data confirming the Y255A mutation in the *xylE* gene of pHISCAT, visualised in SnapGene Viewer. The mutated codon GCC (highlighted in blue, encoding alanine) replaces the wild-type TAC codon (encoding tyrosine). The position number shown above the residue (261) reflects numbering from the start of the entire pHISCAT-encoded fusion protein, which includes the N-terminal 6xHis tag; the corresponding position in the native catechol 2,3-dioxygenase numbering used throughout this report is residue 255 (Y255), as confirmed by the SnapGene tooltip ("Alanine (255)"). The displayed ORF parameters (924 bp, 307 amino acids, 35.1 kDa) match the published values for native catechol 2,3-dioxygenase (Kobayashi et al., 1995), confirming that no insertions, deletions, or other unintended mutations were introduced.

5. Discussion

The aims of this study were to express and purify the 6xHis-tagged catechol 2,3-dioxygenase from the plasmid pHISCAT in *Escherichia coli*, to characterise the purified enzyme, and to investigate the role of the conserved active site residue Tyr255 by introducing a Y255A substitution via site-directed mutagenesis. It was hypothesised that replacing Tyr255 with alanine would reduce or abolish enzyme activity, given the proposed involvement of this residue in substrate deprotonation at the active site (Ishida *et al.*, 2002). The key findings of this study were that (i) the identity of pHISCAT was confirmed by restriction enzyme digestion, (ii) the 6xHis-tagged catechol 2,3-dioxygenase was successfully expressed, purified by Ni-NTA affinity chromatography, and shown to be catalytically active, (iii) the Y255A mutation

reduced but did not completely abolish enzyme activity as assessed by catechol spray assay, and (iv) the mutation was confirmed by DNA sequencing. These results partially support the hypothesis, as the Y255A mutation reduced activity rather than abolishing it entirely.

5.1 Plasmid verification and recombinant protein expression

Restriction enzyme digestion of pHISCAT produced fragment sizes that were consistent with the predicted restriction map derived from the DNA sequence data (Figure 2), confirming that the plasmid supplied was pHISCAT. The minor discrepancies between predicted and observed fragment sizes are attributable to the inherent limitations of estimating fragment size from agarose gel migration, which is influenced by factors such as gel concentration, running conditions, and DNA conformation.

SDS-PAGE analysis of IPTG-induced *E. coli* cell lysates demonstrated the appearance of a protein band at approximately 37 kDa that increased in intensity over the induction time course (Figure 3), consistent with the predicted size of the 6xHis-tagged catechol 2,3-dioxygenase. The wild-type catechol 2,3-dioxygenase subunit has a molecular weight of approximately 35 kDa (Kobayashi *et al.*, 1995), and the addition of the N-terminal 6xHis tag is expected to increase this by approximately 2 kDa. Kobayashi *et al.* (1995) reported that *E. coli* cells carrying the *xylE* gene produced maximum enzyme activity after 10 hours of growth at 42°C in a large-scale 50-litre culture. In the present study, a smaller-scale expression at 37°C with IPTG induction was used, and the increasing band intensity across the 1- and 2-hour time points demonstrated that the *lac* promoter-driven expression system in pHISCAT functioned as expected. *E. coli* remains the most widely used host for recombinant protein production due to its rapid growth, well-characterised genetics, and the availability of numerous expression vectors (Rosano and Ceccarelli, 2014).

The 3-hour IPTG induction sample appeared anomalously faint on the SDS-PAGE gel (Figure 3, Lane 4). Since all protein bands in this lane were reduced in intensity, not just the recombinant protein, this was most likely due to a pipetting or cell lysis error during sample preparation rather than a genuine reduction in protein expression. This interpretation is supported by the fact that IPTG-induced

recombinant protein expression in *E. coli* typically increases or plateaus over a 3-hour induction period and does not decrease (Rosano and Ceccarelli, 2014).

Western blot analysis using Ni-NTA HRP conjugate failed to detect 6xHis-tagged protein on the membrane (Figure 4). It should be noted that the 10 kDa molecular weight marker band was not visible on the membrane in either marker lane, which is consistent with the well-recognised tendency of low molecular weight proteins to transfer poorly to nitrocellulose membranes due to reduced retention during the electrophoretic transfer step. Several factors may have contributed to this outcome, including incomplete transfer of proteins from the gel to the nitrocellulose membrane, insufficient incubation time or incorrect concentration of the Ni-NTA HRP conjugate, or problems during the chromogenic detection step. Despite this failure, the SDS-PAGE results provided sufficient evidence of recombinant protein expression, and the subsequent successful purification on a Ni-NTA column, which specifically binds 6xHis-tagged proteins, provided indirect confirmation that the expressed protein carried a functional 6xHis tag.

5.2 Purification and characterisation of 6xHis-tagged catechol 2,3-dioxygenase

The 6xHis-tagged catechol 2,3-dioxygenase was successfully purified using Ni-NTA affinity chromatography under native conditions, as demonstrated by SDS-PAGE analysis of the purification fractions (Figure 5). The progressive removal of contaminating *E. coli* proteins through the wash steps and the strong enrichment of the ~37 kDa band in the elution fractions confirmed effective purification. Minor contaminating bands were observed in the first and second elution fractions, which is a recognised limitation of Ni-NTA affinity chromatography; *E. coli* naturally produces several metal-binding proteins containing clusters of histidine residues that can interact weakly with the Ni-NTA matrix (Wei *et al.*, 2010). Bands below 37 kDa in the elution fractions may also represent degradation products of the 6xHis-tagged catechol 2,3-dioxygenase that occurred during purification.

During the concentration step using the Amicon Ultra-4 centrifugal filter, only 160 µl of concentrated protein was recovered instead of the expected 400 µl. The initial volume of pooled eluate exceeded the 4 ml capacity of the filter device, requiring an additional centrifugation cycle. It is possible that some concentrated protein was

inadvertently discarded with the flow-through during this additional step. This interpretation is supported by the Bradford assay data, in which the highest usable dilution (1 in 25) gave a mean absorbance at 595 nm of only 0.212, possibly indicating a protein yield lower than anticipated which is consistent with loss of material during the additional centrifugation step.

The specific activity of the purified enzyme was determined to be approximately 3.2 U/mg (where 1 U = 1 µmol 2-hydroxymuconate semialdehyde produced per minute). This is substantially lower than the 536 U/mg reported by Kobayashi *et al.* (1995) for the crystalline holoenzyme purified from a large-scale *E. coli* culture, and also lower than the 75 U/mg reported for the crude extract in the same study. Several factors are likely to account for this difference. First, catechol 2,3-dioxygenase requires a ferrous iron (Fe^{2+}) cofactor at each active site for catalytic activity, and the relationship between iron content and specific activity is well established (Kobayashi *et al.*, 1995). During the small-scale purification used in this study, iron may have been lost or oxidised to the inactive Fe^{3+} form, particularly in the absence of the acetone-containing buffers that Kobayashi *et al.* (1995) used to stabilise the enzyme. Secondly, the Bradford assay may have overestimated total protein concentration if contaminating co-purified proteins were present, which would lower the calculated specific activity. Thirdly, it should be noted that the enzyme activity data presented were representative values rather than the author's own experimental data, as insufficient time was available to complete triplicate assays. This was due to additional time spent on the concentration step and spectrophotometer setup. The decrease in initial rate observed across the three replicate assays (0.176, 0.166, and 0.149 A_{375}/min for assays 1, 2, and 3 respectively) is consistent with the progressive inactivation of diluted catechol 2,3-dioxygenase, a well-documented property of the enzyme that occurs due to the irreversible oxidation of the active site ferrous iron upon dilution (Kobayashi *et al.*, 1995).

5.3 Effect of the Y255A mutation on catechol 2,3-dioxygenase activity

The central finding of this study was that the Y255A mutation reduced but did not abolish catechol 2,3-dioxygenase activity, as assessed by the qualitative catechol spray assay. Sectors containing *E. coli* with potentially mutated pHISCAT showed varying levels of yellow colouration upon spraying with catechol (Figure 8, Table 3).

Sectors 1 to 4 displayed clearly reduced yellow colour intensity (+ to ++) compared to the wild-type pHISCAT positive control (+++), whereas sectors 7 and 8 showed near-wild-type colouration (++ to +++). The pHAT10 negative control showed no yellow colouration, confirming that the colour change was specific to catechol 2,3-dioxygenase activity.

The difference in activity between sectors 1–4 and sectors 7–8 is most readily explained by considering the efficiency of the QuikChange mutagenesis protocol. DpnI digestion selectively destroys methylated parental (wild-type) plasmid DNA, but this digestion is not always 100% complete, and a proportion of colonies are expected to contain unmutated plasmid. The class control data support this interpretation: pWhitescript without DpnI produced too many colonies to count, whereas DpnI-treated pWhitescript produced only 8–12 blue colonies on the 100 µl plates (Table 2), demonstrating effective but not absolute removal of parental DNA. It is therefore likely that sectors 7 and 8 represent colonies containing wild-type pHISCAT that escaped DpnI digestion, while sectors 1–4 contain the Y255A mutant. This conclusion is further supported by the DNA sequencing data, which confirmed the presence of the Y255A mutation (TAC to GCC) in the selected miniprep sample (Figure 10).

The partial reduction in enzyme activity caused by the Y255A mutation is consistent with the proposed role of Tyr255 in the catalytic mechanism of catechol 2,3-dioxygenase. Ishida *et al.* (2002) identified Tyr255 as a conserved active site residue and proposed that it plays an important role, together with His199, in the deprotonation of substrate ligands and the formation of charge-transfer complexes at the Fe(II) centre. The crystal structure determined by Kita *et al.* (1999) showed that the active site of catechol 2,3-dioxygenase is located in a funnel-shaped cavity within the C-terminal domain, with the iron atom coordinated by His153, His214, and Glu265. Tyr255 is not one of these iron-coordinating residues, which explains why the Y255A mutation reduced rather than abolished activity; the iron cofactor can presumably still bind to the active site in the absence of Tyr255, but the catalytic efficiency is impaired because the hydroxyl group of tyrosine is no longer available to participate in substrate deprotonation. Had one of the iron-coordinating residues (His153, His214, or Glu265) been mutated to alanine instead, a complete loss of

activity would have been expected, as the iron cofactor would no longer be able to bind at the active site.

The enzyme's broad substrate specificity reported by Cerdan *et al.* (1994), including its ability to cleave catechol, 3-methylcatechol, and 4-methylcatechol, suggests that the active site must accommodate substrates of varying size while maintaining the correct geometry for extradiol cleavage. The partial retention of activity in the Y255A mutant indicates that while Tyr255 contributes to the catalytic process, it is not absolutely essential, and other residues or structural features of the active site cavity may partially compensate for its loss. Further quantitative characterisation of the Y255A mutant, such as determination of K_m and k_{cat} values, would be required to fully understand how this mutation affects substrate binding and catalytic turnover. Ishida *et al.* (2002) reported K_m values of 1.87 μM for catechol and 7.5 μM for O_2, with a k_{cat} of 278 s^{-1} for the wild-type enzyme; comparison of these parameters with those of the Y255A mutant would reveal whether the mutation primarily affects substrate binding affinity, catalytic rate, or both.

5.4 Critical appraisal of experimental limitations

Several experimental limitations should be acknowledged when interpreting the results of this study. The Western blot failed to detect any 6xHis-tagged protein, which prevented immunological confirmation of recombinant protein expression; however, the SDS-PAGE data and the successful Ni-NTA purification provided alternative evidence that the protein was expressed and carried a functional 6xHis tag.

During the site-directed mutagenesis protocol, step C of the transformation procedure (a brief incubation step) was inadvertently omitted. Despite this, colonies were obtained at reasonable numbers, indicating that the transformation was not critically compromised. Two sectors (5 and 6) failed to grow when re-streaked, which was most likely due to those particular colonies not being picked up successfully rather than being a consequence of the skipped transformation step, since other sectors from the same transformation grew normally.

The catechol spray assay used to assess the effect of the Y255A mutation is a qualitative method that provides only a semi-quantitative comparison of enzyme activity between sectors. Ideally, the mutant enzyme would have been purified and its specific activity determined quantitatively under controlled conditions, allowing direct comparison of kinetic parameters with the wild-type enzyme. Similarly, whilst the DNA sequencing confirmed the Y255A mutation in one miniprep sample, it would have been informative to sequence additional colonies, particularly from sectors 7 and 8, to confirm whether these contained wild-type plasmid as hypothesised.

The enzyme activity data presented in this report were representative values, as the author was unable to complete triplicate assays within the available time. This limits the statistical confidence that can be placed on the specific activity calculation. Nevertheless, the data are internally consistent and demonstrate the expected enzyme kinetics profile, with a rapid initial rate followed by a plateau as substrate becomes limiting.

5.5 Conclusions

In conclusion, this study successfully expressed and purified 6xHis-tagged catechol 2,3-dioxygenase from pHISCAT in *E. coli* and demonstrated that the enzyme was catalytically active. Site-directed mutagenesis was used to introduce a Y255A substitution, which was confirmed by DNA sequencing. The Y255A mutation reduced but did not abolish catechol 2,3-dioxygenase activity as assessed by catechol spray assay, partially supporting the hypothesis that Tyr255 is important for catalytic function. These findings are consistent with the proposed role of Tyr255 in substrate deprotonation at the active site (Ishida *et al.*, 2002) and with the crystal structure data showing that Tyr255 is a conserved active site residue located near but not directly coordinating the catalytic iron (Kita *et al.*, 1999). The results suggest that Tyr255 contributes to but is not solely responsible for the catalytic activity of catechol 2,3-dioxygenase, and that other active site residues or structural features may partially compensate for the loss of the tyrosine hydroxyl group.

6. Future Work

Several further experiments could be conducted to build upon the findings of this study and improve understanding of the structure and function of catechol 2,3-

dioxygenase. The most immediate next step would be to purify the Y255A mutant enzyme using the same Ni-NTA affinity chromatography protocol used for the wild-type protein and to determine its specific activity, K_m, and k_{cat} values under controlled conditions. This would allow a direct quantitative comparison with the wild-type kinetic parameters reported by Ishida *et al.* (2002) and Kobayashi *et al.* (1995) and would reveal whether the Y255A mutation primarily affects substrate binding affinity, catalytic turnover rate, or both. Additionally, site-directed mutagenesis of other conserved active site residues, particularly His199 (which Ishida *et al.* (2002) proposed works together with Tyr255 in substrate deprotonation), as well as the iron-coordinating residues His153, His214, and Glu265, would provide a more comprehensive map of structure-function relationships at the active site.

Structural characterisation of the Y255A mutant by X-ray crystallography or homology modelling could reveal whether the mutation causes any local conformational changes at the active site that might explain the reduced catalytic activity. It would also be valuable to test the substrate specificity of the Y255A mutant against a panel of substituted catechols, as described by Cerdan *et al.* (1994), to determine whether the mutation differentially affects activity towards different substrates. From an applied perspective, understanding which residues are essential for catechol 2,3-dioxygenase activity has implications for the engineering of enzymes with improved stability or altered substrate specificity for use in bioremediation of aromatic pollutants, an area of increasing environmental importance.

7. References

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8. Appendix

This appendix provides the raw experimental data, supporting tables, and supplementary calculations underlying the results presented in the main report.

Table 1. Number of colonies obtained on LB-amp agar plates following site-directed mutagenesis of pHISCAT (Y255A) and transformation into XL1-Blue supercompetent cells. Plates 1–4 received 100 µl of transformation mixture; plates 5–6 received 20 µl.

Plate	Volume plated (µl)	Number of colonies
1	100	70
2	100	79
3	100	81
4	100	56
5	20	4
6	20	21

Table 2. Class control colony counts on LB-amp X-gal IPTG agar. The pWhitescript no DpnI control demonstrates the high background of parental DNA. The pWhitescript plus DpnI control demonstrates that DpnI digestion effectively removes parental methylated DNA. The *E. coli*/pUC18 mixture serves as a transformation efficiency control.

LB amp X-gal IPTG agar	**Plate 1 (100 µl)**	**Plate 2 (100 µl)**	**Plate 3 (10 µl)**	**Plate 4 (10 µl)**
pWhitescript no DpnI control	White: too many to count Blue: 0	White: too many to count Blue: 0	White: 920 Blue: 0	White: 987 Blue: 0
pWhitescript plus DpnI control	White: 0 Blue: 8	White: 0 Blue: 12	White: 0 Blue: 1	White: 0 Blue: 3
E. coli/pUC18 mixture	Too many to count	Too many to count	560	420

Table 3. Qualitative scoring of yellow colour development upon spraying IPTG-induced *E. coli* sector streaks with 100 mM catechol. Scoring scale: − = no yellow colouration; + = faint yellow; ++ = moderate yellow; +++ = strong yellow.

Sector	**Yellow colour at 0 min**	**Yellow colour at 5 min**	**Yellow colour at 10 min**
1	−	+	+
2	−	+	++
3	−	+	+
4	−	+	++
5	No growth	No growth	No growth
6	No growth	No growth	No growth
7	−	++	++
8	−	++	+++
E. coli pHAT10 (negative control)	−	−	−
E. coli pHISCAT (positive control)	−	+++	+++

Table 4. Bovine serum albumin (BSA) standard curve raw data for the Bradford protein assay. Each BSA concentration was assayed in duplicate (Sample A and Sample B), and the absorbance at 595 nm was measured against a blank (0 µg BSA). Mean absorbance values were used to plot the standard curve (Figure 6).

BSA amount (µg)	**Absorbance 595 nm Sample A**	**Absorbance 595 nm Sample B**	**Mean absorbance 595 nm**
0 (Blank)	0.000	0.000	0.000
2.5	0.168	0.179	0.174
5.0	0.350	0.337	0.344
7.5	0.476	0.480	0.478
10.0	0.532	0.570	0.551
12.5	0.692	0.670	0.681
15.0	0.759	0.783	0.771
17.5	0.822	0.841	0.832
20.0	0.924	0.896	0.910

Table 5. Bradford assay results for the dilution series of purified 6xHis-tagged catechol 2,3-dioxygenase. Only the 1 in 25 dilution produced a mean absorbance value within the linear range of the BSA standard curve (Table 4) and was therefore used to back-calculate the protein concentration of the concentrated stock.

Dilution of purified 6xHis-tagged catechol 2,3-dioxygenase	**Absorbance 595 nm Sample A**	**Absorbance 595 nm Sample B**	**Mean absorbance 595 nm**	**Within standard curve range?**
Concentrated (undiluted)	2.361	2.423	2.392	Too high
1 in 5	0.844	1.080	0.962	Too high
1 in 25	0.247	0.177	0.212	**Yes**
1 in 125	0.026	0.030	0.028	Too low
1 in 625	−0.007	−0.000	~0.000	Too low
1 in 3125	−0.007	0.004	~0.000	Too low

Table 6. Raw absorbance data at 375 nm from three replicate enzyme activity assays of purified 6xHis-tagged catechol 2,3-dioxygenase (1 in 625 dilution). Note: these data are representative results provided by the module academic with permission and are not the author's own experimental data, due to the time constraints described in the main report.

Time (min)	Absorbance 375 nm Assay 1	Absorbance 375 nm Assay 2	Absorbance 375 nm Assay 3
0	0.352	0.282	0.208
1	0.528	0.448	0.357
2	0.694	0.583	0.483
3	0.811	0.674	0.560
4	0.892	0.762	0.604
5	0.945	0.799	0.633
6	0.969	0.823	0.647
7	0.978	0.840	0.668
8	0.988	0.847	0.683
9	0.998	0.857	0.694
10	0.998	0.862	0.702

Table 7. Summary of enzyme activity calculations derived from the raw absorbance data in Table 6. Initial rates were calculated from the linear portion of each assay (0–3 minutes). Rate of product formation was calculated using the Beer-Lambert law (ε = 36,000 M^{-1} cm^{-1} for 2-hydroxymuconate semialdehyde) for a 1 ml reaction volume. Specific activity was calculated as the rate of product formation divided by the mass of enzyme in the assay (~1.5 µg per assay based on the protein concentration determined from Table 5).

Parameter	Assay 1	Assay 2	Assay 3	Mean
Initial rate (ΔA_{375}/min)	0.176	0.166	0.149	0.164
Rate of product formation (µmol/min)	0.0049	0.0046	0.0041	0.0045
Specific activity (µmol/min/mg = U/mg)	3.20	3.07	2.70	2.99

Determination of protein concentration of concentrated 6xHis-tagged catechol 2,3-dioxygenase from the BSA standard curve:

The 1 in 25 dilution gave a mean absorbance at 595 nm of 0.212. From the BSA standard curve (Figure 6, Table 4), this absorbance corresponds to approximately 3.0 µg BSA-equivalent protein in the 10 µl assay volume, equivalent to a protein concentration of 0.30 mg/ml in the 1 in 25 diluted sample. Multiplying by the dilution factor of 25, the concentration of the concentrated 6xHis-tagged catechol 2,3-dioxygenase stock is approximately 7.5 mg/ml.

Calculation of specific activity (Assay 1, worked example):

Initial rate (ΔA_{375}/min) from the linear region (0–3 min) of Assay 1 = 0.176 min^{-1}.

Using the Beer-Lambert law ($A = \varepsilon cl$) where ε = 36,000 M^{-1} cm^{-1} and l = 1 cm:

Rate (M/min) = 0.176 / 36,000 = 4.89×10^{-6} M/min

For a 1 ml reaction volume:

Rate (µmol/min) = 4.89×10^{-6} mol/L × 0.001 L × 10^{6} µmol/mol = 0.00489 µmol/min

The 50 µl of 1 in 625 diluted enzyme added to the reaction contained approximately:

(7.5 mg/ml ÷ 625) × 0.050 ml = 0.0006 mg = 0.0015 mg (using worksheet protein values)

Specific activity = 0.00489 µmol/min ÷ 0.0015 mg ≈ **3.2 µmol/min/mg (= 3.2 U/mg)**